



"The agent that has to **prove** its trade."

FRANTAL COMPANY

The Problem with LLMs in Trading

VULNERABILITIES

Mathematical Hallucination

Standard generative LLMs are statistical language models. When applied directly to risk modeling, options pricing (Black-Scholes, Greeks), or expected-value (EV) calculations:

- ✗ They tend to invent probabilities that sound plausible but are false.
- ✗ They offer no guarantee of deterministic numerical precision.
- ✗ They make subtle calculation errors with catastrophic financial impact.

Execution Uncertainty

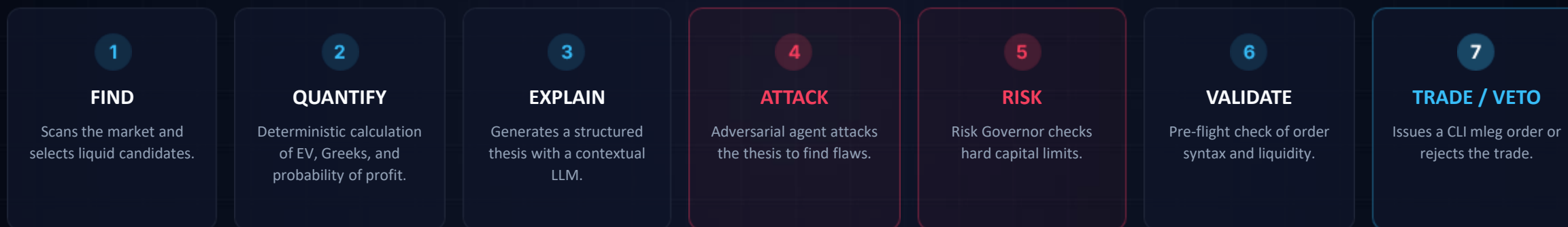
Submitting trading orders based solely on generative outputs, without rigorous verification layers, introduces unacceptable risks:

- ⚠ **Lack of Hard Limits:** There's no way to guarantee an LLM respects capital limits without external safeguards.
- ⚠ **Confirmation Bias:** Standalone models rationalize high-risk trades while ignoring adverse scenarios.

Solution: The ORION Decision Pipeline

WORKFLOW

To eliminate generative risk, ORION implements a rigorous **chain-of-proof** workflow before any execution:



Key Differentiator: A trade only executes if it clears the adversarial filter and respects the Risk Governor's exact mathematical limits.

Hybrid Intelligence: Reasoning vs. Precision

ARCHITECTURE

Dimension	Deterministic Engine (Quant)	Generative Engine (LLM)
Responsibility	Pricing, EV, Greeks, Liquidity, Risk Limits	Thesis Interpretation, Red-Teaming, Synthesis
Output Nature	Exact math (Python / SciPy / QuantLib)	Structured Natural Language (JSON)
Precision Guarantee	100% reproducible, zero margin of error	Probabilistic / Subject to critical reasoning
Decision Authority	ABSOLUTE VETO (Risk Governor Override)	Proposal & Adversarial Rejection

ORION combines the best of both worlds: the analytical flexibility of an LLM with the unbreakable mathematical discipline of quantitative code.

Risk Gates & Governance

PROTECTION



Risk Governor

Independent deterministic module. Enforces hard Max Loss limits per trade and daily portfolio exposure limits.



Position Sizing

Dynamic position sizing via quantitative criteria (Kelly / Fixed Fractional), preventing overallocation.



Validator Engine

Pre-flight check that verifies the technical validity of option contracts, reasonable bid-ask spread, and account balance.



Safe Environment: Paper Trading Only

All system executions are strictly isolated in a simulated environment (Paper Trading Sandbox), guaranteeing zero capital risk during validation testing.

Alpaca Engine Integration

TECH STACK

\$100k

Paper Trading Account

Simulated capital under the agent's active management.

REST API

Market Data Feed

Real-time market data and options chains.

CLI / mleg

Order Execution

Native support for complex multi-leg structures.

> CLI Execution Example (mleg)

```
$ orion trade execute --symbol SPY --strategy "Long Put" --legs "BUY 759 Put 2026-09-11" --contracts 3 --account PA3B2R5S7KPW
[SUCCESS] Pre-flight validation passed. Order submitted to Alpaca Paper Trading
Order ID: ord_8f92a10b-4c2d | Status: Filled @ $3.06
debit
```



Results & Real-Time VETO

DASHBOARD

TOP OPPORTUNITY — SPY

Long Put

2026-09-11 • 9DTE

Alpha Score

70

Risk Score

45

Liquidity

83

Prob. of Profit

27%

EV \$27,78 R:R 3.05 Max Loss \$301,5 Max Profit ∞

AI THESIS **WATCH** DETERMINISTIC • CONF 60

SPY Long Put aligns with a BULLISH regime. The structure carries positive expected value (\$27.78) with a 27% probability of profit and a 3.05 reward-to-risk over 9 days.

- Alpha Score 75/100
- Expected value \$27.78
- Probability of profit 27%
- Risk/reward 3.05

ADVERSARIAL CHALLENGE **REJECT** PENALTY -4.6

The Long Put may be overrated: probability of profit is only 27%. With 9 days to expiration, the edge depends on timing and could erode if the BULLISH read is wrong.

- Probability of profit is only 27%
- Risk score 45 is elevated
- Long premium is exposed to IV contraction and time decay

NO TRADE

Adversarial Agent rejected the trade

Real Decision: "NO TRADE"

Even with an Alpha Score of 70, the **Adversarial Agent** flagged a low probability of profit (27%) and triggered the Risk Governor's **VETO**.

Test Coverage

Over **93 automated tests** passed, confirming the robustness of the risk pipeline and CLI execution via Alpaca.

i The dashboard shows the system autonomously declining the trade to protect simulated capital.

ORION

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Account Details

Account ID: PA3B2R5S7KPW

Mode: Paper Trading Only

Team: FranTal Company

Code & Repository

Access the project repository to view the documentation and the 93 tests:

<https://github.com/frantal/orion>

Questions & Answers

Thank you for your attention.